

Andrey Shcherbinin

**Real Time Visualization Design Engineer, Procedural Technical Artist,
Unreal Engine and Houdini Specialist**

(980) 844-0771 gridtraveler@gmail.com U.S. Citizen

Education

Bachelor of Science - Computer Information Systems / Spanish Language (professional proficiency)
Emporia State University - Emporia, Kansas. 2002 / + Aviation Maintenance + Logistics

LinkedIn: <https://www.linkedin.com/in/gridtraveler>

Digital Works: <https://www.artstation.com/gridtraveler>

Sound Design: <https://soundcloud.com/gridtraveler>

Summary

Real time visualization design engineer and procedural technical artist with 10 plus years of experience creating high quality 3D, VFX, XR, simulation, virtual production, UI and UX, and cinematic content for automotive, entertainment, broadcast, live events, industrial, and emerging technology projects. Strong generalist with deep production experience across Unreal Engine, Houdini, Cinema 4D, Blender, NVIDIA Omniverse, OpenUSD, Niagara, PCG, simulation workflows, procedural systems, look development, motion design, technical art, and real time rendering. Known for bridging creative design, technical execution, rapid prototyping, pipeline problem solving, visual storytelling, stakeholder communication, and production delivery across complex multidisciplinary teams.

Work Experience

Volvo Group - North America

Real-Time 3D Automotive Visualization Engineer

July 2024 - July 2025

Collaborated with vehicle designers and engineers to enhance the design process through the development of high-fidelity digital twins of vehicles and real-time environments in virtual reality (VR). Developed interactive MR and VR apps in the Open XR C++ environment working with Varjo XR4 headset and ultraleap for hand tracking.

Designed and implemented interactive VR experiences applying UX principles for immersive environments that empower the team to review materials, fabrics, shapes, and overall vehicle aesthetics in real time, fostering a more collaborative and efficient design review process.

Implemented multi user VR collaborative environments for vehicle design and configurations evaluation.

Pioneered real-time rendering techniques and optimization strategies to ensure seamless performance and exceptional visual quality of VR simulations, enhancing the team's ability to make informed decisions.

Building photo realistic environments for real time photorealistic vehicle digital twins interactions for early vehicle design validation.

Electronic Countermeasures Deloitte - Los Angeles, California

Unreal Engine Tech Lead Generalist / Real-Time UE5 Niagara VFX Designer

Jul 2024 - Dec 2024

Built C++ responsive procedural real-time DMX lighting systems and shaders for Leon Bridges performance.

Designed mission-critical Niagara VFX systems for Deloitte's flagship event at the Las Vegas Sphere, enabling real-time storytelling for 20,000+ attendees and 5M+ virtual viewers across 15+ countries.
Pioneered adaptive volumetric rendering workflows to synchronize multi-sensory effects (light, particle, sound) with keynote narratives, enhancing audience engagement by 40% post-implementation.
Engineered GPU-optimized parameter interfaces for non-technical teams, reducing VFX iteration time by 55% while maintaining stable performance.
Led cross-functional collaboration with teams to align real-time systems with Deloitte's global brand standards, ensuring seamless integration.

Raw Cereal - Los Angeles, California

Unreal Engine Tech Lead Generalist / Real-Time UE5 Niagara VFX Designer

December 2022 - March 2024

Developed C++ interactive visual systems for the Morgan Wallen world tour concerts that included real-time Unreal Engine environments and Niagara systems for the stage I.e.d. Volumes. Created interactive Niagara systems for Disguise that include both sound reactive shaders and sound driven Niagara systems.

Possible - Los Angeles, California

Unreal Engine Developer Technical Lead Artist Real-time Generalist

July 2023 - January 2024

Developed VFX including Niagara Systems and custom performant shaders for a mixed reality adaptation in a large scale live tv production. Worlds 2024 League of Legends, South Korea event. C++

Xite Labs - Los Angeles, California

Unreal Engine Cinema 4D VFX Generalist

July 2023 - February 2024

Designed VFX & 3D visualizations in Cinema 4D and Unreal Engine including realtime VFX with C++ Blueprints, organic motion design, 3D environments lighting and color, animations for large scale live events - Hilton Keynote, Las Vegas CES - Walmart, Formula E.

Biolumina - NYC, New York

Biomedical 3D Generalist

February 2022 - July 2023

Created biomedical 3D visualizations including intracapillary and at nano scale.

Trilith Studios NEP - Atlanta, Georgia

Real-Time Virtual Production Unreal Engine 5 Generalist

October 2022 - December 2022

Operation of Unreal Engine within I.e.d.volume real-time ecosystem. Mostly training.

The Weather Channel - Atlanta, Georgia

Sr. Environment Artist | Unreal Engine Generalist / Real-time Virtual Production 24/7 Virtual Weather

July 2022 - March 2023

Advanced the state of real-time rendered immersive experiences, striving for highest possible visual fidelity and artistic innovation. Built assets and C++ Blueprints environment
Worked closely with the design leads to conceptualize, model, dress and light real-time worlds based on related concept art and photographic reference.

Katy Perry Play Las Vegas - Hi From The Future - Vegas / New York

Unreal Engine 5 Generalist / 3D Character Artist Animation / Simulations & VFXs Generalist

October 2021 - December 2021

Worked with an amazing team to help with production of imax size high fidelity visuals, character animations, vfx and simulations in UE5 Unreal Engine 5.

Worked closely with key team members to transfer storyboard sketches into high fidelity Unreal Engine 5 UE5 real-time 3D environments, helped to setup characters in Unreal Engine 5 UE5, apply motion capture data, setup grooms/hair for characters, setup hundreds of camera shots and fine tuned each camera for best aperture/fstop/iso/shutter/focus settings to work best with the lighting, helped to set up natural lighting to accentuate key elements and to strengthen overall composition while maintaining original pantone palette color scheme, helped to align characters animations and vfx timing to the soundtrack.

Created accurate softbody/cloth dynamic simulations. Created real-time VFX with UE5 Niagara for sparks, water, fire to work in conjunction with scene elements and character animations. Setup all the shots in sequencer and worked with UE5 movie render que to rapidly deliver hd cinematics iterations. Helped to identify key UE5 console variables/commands for rendering the highest visual quality product possible.

BSI Solutions - Charlotte, North Carolina

VR Technical Artist - Unity & Unreal Engine / 3D Medical Animator / UI UX VFX Generalist / VR Environments Designer

October 2020 - October 2021

Created and validated 3D content with Maya and Blender against technical and performance guidelines.

Developed AR and VR apps for education.

Architected lean and performant VR scenes for mobile headsets - Pico and Quest.

Created organic and non-organic optimized 3D geometry ranging from nanoscale medical bio VR assets to large scale VR environments.

Lighting and custom shaders for real-time VR in C#

Created complex event driven animation timelines in Unity C#

Duke Energy - Charlotte, North Carolina

VR UI UX VFX 3D Generalist, Software Architect / Unreal Technical Artist

August 2018 - September 2020

Developed AR and VR apps for training, gaming and education.

Innovated workflows to maximize visual quality and fps in Unreal Engine.

Designed immersive VR experiences and optimized assets, textures, shaders in Unreal engine C++ Blueprints.

Built Unreal blueprints for interaction in VR.

Solved complex VR issues to create the next generation of seamless high fps immersion. Designed optimized environments, VFX and assets to run on Oculus Quest.

Designed and tested VR content on Quest, Vive and Rift headset.

Live Nation Entertainment - New York, New York

Freelance VFX VR Cinema 4D Generalist, 3D Animator.

June 2018 - July 2018

Created 3D assets and integrated with live footage, 3D visualizations for Cannes film festival short.

Oath - New York, New York

3D Generalist, Motion Designer, VFX Artist, 3D Animator

December 2015 - July 2018

Designed and animated daily live content graphics for the What to Watch show, both still and animated. Creating style frames and animations for new show segments, and packages. Designed VFX. Creating templates. Conceptualized and delivered key 2D and 3D animations.

Built highly time sensitive daily digital assets - animations, motion graphics, illustrations.

Created animations and VFX including opening titles & lower thirds.

Conceptualizing, 3D Modeling and Animation, motion graphics, rotoscoping, typography, animation, motion/camera tracking, keying, lighting, texturing, data visualization.

Ripple Collective - New York, New York

VFX VR Cinema 4D Generalist, Animator, 3D VR Environments Architect

March 2017 - June 2018

Conceptualized, built, lighted, textured and animated 3D environments to visualize blockchain technology. Adobe After Effects, green screen, Cinema 4D, Premiere, video editing, compositing, VFX, 360 VR design and editing. Created 360 3D environments including Mars surface, created space station, created intergalactic warp speed traveling teleportation modules for 360 VR experience. Created 3D data visualization for 360 VR environments. Data viz in VR. Created VR content. Conceptualized and built 3D environments, 3D VR Data Visualization, 360 VR 3D sets, 3D assets, VFX, lighting, Arnold shader networks, Cinema 4D animations and transitions.

Viacom - New York, New York

VFX Artist, 3D Animator, Motion Graphics Designer - Comedy Central

August 2017 - December 2017

Created narrative graphics, 3D animations and informational motion graphics in a high pressure on-air deadline-oriented environment. Conceptualizing and building 3D animations, including 3d modeling, lighting, texturing, VFX and motion graphics for The President Show

Working in an extremely fast paced live tv environment making critical creative direction decisions.

The Molecule VFX - New York, New York

3D Generalist, Cinema 4D & Look Development Artist

January 2017 - February 2017

Created the look for photo realistic delivery constructing advanced Arnold Renderer shader networks. Ensured look and feel consistency and built precise animations to deliver the story.

Created 3D scenes for further processing through render farm. Composited, extracted and color corrected EXRs while keeping files and structures organized. Worked with the most amazing talented and kind people in the world that are a part of The Molecule VFX.

Core Augmented Reality Education - New York, New York

Creative Director / Virtual Reality UI UX Designer / 3D Sculptor and Animator / VR Artist / VFX Designer / Cinema 4D Generalist / Unity 3D Animator / VR 3D Environments Artist / Lighting TD / Textures TD

March 2016 - March 2017

Built next generation futuristic 3D environments, characters, VR Virtual Reality guide crafts, futuristic technology for HTC Vive and Oculus. Helped to shape Augmented Reality apps.

Constructed full scenes in real-time engine Unity3D C# environment including texturing assets in Substance Painter, Vuforia integration.

Scaled assets from PC platforms (Rift) to mobile systems (Gear VR) while maintaining visual fidelity.

Helped a VR AR startup with the visual, creative and aesthetic side of business. Helped to create next generation immersive AR VR experiences.

Created a brand new brand presence on the web. Designed this responsive website from scratch (www.care4d.com) Work included art direction, UI UX design, entire look and feel for both innovation and education websites including the landing page.

Created brand new UI for Care4D mobile app. Created new UI for education and agency websites, created custom art assets and new look and feel for both. Created AR and VR concepts for implementation in Unity engine. Provided creative direction to Unity developers.

Designed this responsive website from scratch as an additional project besides VR projects (www.care4d.com)

Soho Experiential - New York, New York

3D Generalist & Motion Graphics Designer, Cinema 4D Generalist

April 2016 - October 2017

Conceptualizing, 3D Modeling and Animation, motion graphics, rotoscoping, typography, animation, motion/camera tracking, keying, lighting, texturing, architectural visualization, data visualization.

Designing 3D animations and visual effects for Fortune 500 Brands / TV Networks

Bravo Network / Below Deck show - 3D animations, camera tracking, visual effects.

The Macallan whisky brand - 3D animations, visual effects.

McKinsey & Company - New York, New York

Freelance Animator & Motion Graphics Designer, Cinema 4D Generalist

April 2015 - 2017

Designed branded animations & motion graphics. Designed and edited 360 VR content.

High 5 Games - New York, New York

3D Generalist / Motion Graphics Designer / Visual Effects Engineer / Cinema 4D Generalist

July 2013 - July 2015

Took ideas and still art and brought them to life by creatively animating and applying visual effects. Excelled at 3D Modeling, Texturing, Lighting, Dynamics, Animation, Effects, Cinema 4D, Maya, ZBrush, Vue XStream, After Effects. Conducted R&D and presented new concepts. Experienced in Shotgun and Jira environments.

Created animations that included conversion of 2D assets into 3D by sculpting and texture projections. Built, textured and lighted digital oceans, landscapes, architecture, people, creatures, atmospheres, trees and vegetation, natural phenomenon, particles and special effects, dynamics. Worked with live footage rotoscoping, green screen keying, motion tracking, color correction, color grading. Directed actors.

Freelance - Brooklyn, New York, Wichita, Kansas

Animator / 3D Character Artist / Motion Graphics Designer / Creative Director / UI UX Web Designer

January 2002 - 2013

3D Modeling, Animation, Lighting, Rigging, Simulations including medical & human anatomy. Motion Graphics Design for Web,TV, Billboards, Projection Mapping, Interfaces and VR. Product Design including custom 3D jewelry. Projection Mapping & Sound Engineering. Custom Virtual Reality Interactive Environments. Traditional & Digital Illustration & Animation EDM Music Production. Continuous learning included Cinema 4D, 3ds Max, Blender and Maya softwares and related plugins. Redesigned Pizza Hut's BOOK IT! Program websites including new look and feel, navigation, summer landing pages, created interactive content that led to an increase in traffic. Created user centered interactive digital content that led to program's growth. Create animated video content that lead to increase in audience participation and increase in sales. Generated and turned new ideas into a digital format enabling customer growth. Designed wireframes, prototypes, flow diagrams for financial as well as mainstream applications including ui for children. Applied design principles, balance, light, color and

timing. Designed interactive experiences & web banners. Designed for infographics and data visualization. Analyzed the audience and developed engaging digital content for kids & adults. Interpreted large sets of information into visual and interactive formats. Photographed, filmed, edited, composited + audio edited. Created effective animations (hand drawn, 2d + 3d) Created visual content for various audiences including children. Worked with Kendo UI and Wijmo frameworks. Produced scientific, medical, pharmaceutical animations that included 3d modeling, lighting, texturing, animation, rendering, video compositing, sound design & audio compositing. Worked with HTML5, CSS3, SASS, LESS, Grids, Javascript, JQuery. Sound engineered music including fractal edm. Translated ideas into easy to understand and action ready designs or animations. Created environments, prefabs, 3d objects and textures, prepared states and events for interactivity within the Unity environment.

CH Robinson Worldwide - Wichita, Kansas
Sales executive, 3PL, Logistics Management.
January 2002 - 2008

Serviced and grew relationships with multiple assigned accounts; primarily focused on the operational execution of the business of C.H. Robinson. Applied standard operating procedures, key performance indicators, and other account metrics. Collaborated with an account management team, supervisors and managers in forecasting, business planning, and strategy. Participated in face-to-face meetings. Proactively developed solutions for service improvement and operational enhancements. Independently managed, serviced and cultivated customer relationships. Helped develop, participate in and lead business reviews and other customer meetings. Collaborated with the team for pricing decisions, the selection of suppliers, and timing of shipments. Engaged in budget preparation and led account activities toward budgeted goals. Sold core products, modes and services. Independently generated new sales revenue by selling nationwide. Created quotes for both customers and carriers. Invoiced customers. Effectively communicated and negotiated internally and externally. Served multiple customers on daily bases. Sold services to senior executives. Stayed competitive, self-motivated. Stayed focused and paid attention to detail for increased accuracy. Multi-tasked in a highly stressful fast paced environment. Worked within a team to support a team. Made time sensitive monetary decisions. Handled incredible amounts of stress - sometimes selling to dozens of accounts at once in real time. Contributed to growth, sales relationship building, and increased financial acumen. Built lucrative lasting business relationships. Dedicated time and followed through. Persistently analyzed complex data sets and problems solved under high pressure. Successfully implemented technology and systems. Proven success collaborating with others to create customized logistics solutions.

Core skills

Real time visualization, technical art, procedural content creation, 3D generalist production, Unreal Engine development, Houdini procedural workflows, VFX systems, real time VFX, simulation design, cinematic rendering, motion design, XR visualization, VR development, MR development, AR prototyping, virtual production, digital twins, industrial visualization, automotive visualization, broadcast graphics, live event visuals, UI and UX for real time systems, interactive prototypes, look development, lighting, shading, materials, asset optimization, environment creation, hard surface modeling, vehicle visualization, procedural environments, particles, fluids, crowds, destruction, data driven visuals, real time compositing, performance optimization, pipeline automation, tool building, cross functional collaboration, stakeholder presentation, creative direction, visual problem solving, rapid prototyping, research and development.

Unreal Engine skills

Unreal Engine 5, Blueprint systems, Niagara VFX, PCG, Lumen, Nanite, Chaos, MetaSounds, Sequencer, Control Rig, materials, material instances, dynamic materials, post process, cinematic cameras, lighting, real time rendering, optimization, level design, environment assembly, virtual production workflows, XR workflows, OpenXR, Cesium, USD import, Datasmith, asset pipelines, UI widgets, interactive prototypes, simulation visualization, technical troubleshooting.

Houdini skills

Houdini FX, procedural modeling, SOP workflows, VOP workflows, VEX fundamentals, Solaris, Karma, USD workflows, terrain systems, HeightFields, particles, POP networks, FLIP fluids, whitewater, oceans, vellum, RBD, crowds, agents, simulation caching, procedural scattering, attribute workflows, point clouds, curve systems, instancing, digital assets, procedural environments, motion graphics systems, export pipelines for Unreal Engine.

3D and rendering skills

Modeling, hard surface modeling, environment art, set extension, UV layout, texturing, look development, shading, lighting, camera animation, cinematic layout, photoreal rendering, stylized rendering, product visualization, automotive rendering, architectural visualization, material creation, displacement, normal maps, reflections, ray tracing, HDRI lighting, render optimization, compositing passes, final image polish.

Motion design and VFX skills

Motion graphics, procedural animation, particle systems, real time effects, cinematic effects, energy effects, weather effects, smoke, fire, water, impact effects, holographic interfaces, audio reactive visuals, broadcast graphics, title design, event screen content, projection visuals, LED visuals, abstract visual systems, animated storytelling, visual rhythm, design systems for motion.

XR and simulation skills

Virtual reality, mixed reality, augmented reality, real time simulation, design review environments, automotive XR validation, industrial training visualization, interactive 3D prototypes, immersive experience design, digital twin environments, headset workflows, Varjo workflows, OpenXR workflows, spatial interaction concepts, virtual cameras, real time interface design.

UI and UX skills

Real time UI and UX, Figma, interactive prototyping, interface layout, screen graphics, HUD design, HMI visualization, dashboard concepts, visual hierarchy, usability minded design, motion UI, 3D interface design, immersive UI, Unreal UI, prototype systems, design communication, user flow thinking, product visualization interfaces.

Pipeline and technical skills

OpenUSD, NVIDIA Omniverse, asset pipeline design, naming conventions, version conscious workflows, import and export troubleshooting, FBX, USD, GLB, Alembic, texture optimization, mesh optimization, LOD thinking, shader troubleshooting, procedural automation, Python fundamentals, Blueprint logic, data driven content, real time performance tuning, scene organization, render troubleshooting, production documentation.

Software proficiencies

Unreal Engine, Houdini FX, Cinema 4D, Blender, NVIDIA Omniverse, OpenUSD, USD Composer, Solaris, Karma, Redshift, Octane, After Effects, DaVinci Resolve, Photoshop, Illustrator, Figma, Substance Painter, Substance Designer, ZBrush, Marvelous Designer, Nuke fundamentals, TouchDesigner, ComfyUI, Git fundamentals, Jira, Trello, Perforce fundamentals, Plastic SCM fundamentals, Microsoft Teams, Google Workspace.

AI and emerging technology skills

AI assisted concept development, AI image workflows, ComfyUI, Stable Diffusion workflows, reference generation, moodboard development, visual exploration, upscaling, creative ideation, generative design support, AI enhanced production planning, image to concept workflows, AI assisted storytelling, responsible creative use of AI.